Prathit Sharma

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Mechanical Engineering undergraduate at IIPE working across product design, engineering analysis and consulting. Hands-on exposure to aerospace hardware manufacturing, rocket-propulsion research and hydropower O&M, with MATLAB and Python models behind design decisions and a consulting track record through 180 Degrees Consulting and national case competitions.

SKILLS

- **Engineering Domains:** Fluid Mechanics, Thermodynamics, Turbomachinery, Systems Engineering, Trade Studies, Mechanism Selection, Euler Buckling Analysis, Failure Mode Diagnosis, Pneumatic System Sizing, Thermochemical Analysis.
- **Design & CAD:** SolidWorks (Parametric 3D CAD, Assemblies), Autodesk Fusion 360, AutoCAD, Section/Exploded Views, BOM Preparation.
- **Programming & Computation:** MATLAB (Parametric Sweeps, Iterative Solvers), Python (pandas, SciPy), MS Excel.
- **Analysis Tools:** NASA CEA, Abaqus (FEA, basic), MS Excel, LaTeX, Git.
- **Quality & Business:** Six Sigma (DMAIC), QMS & Technical Documentation, Financial Modelling (NPV, Payback), Monte Carlo Risk Analysis.

INTERNSHIPS

SEC Industries Private Limited **May 2026 – June 2026**
Engineering Intern *Hyderabad, India*

– Rotated through Design, Quality Control, Fabrication, Machine Shop and Planning at an aerospace hardware manufacturer, seeing how a design is taken to manufacture.
– Worked with the organisation's Quality Management System (QMS), gaining hands-on exposure to technical documentation, traceability records and compliance workflows.
– Conducted preliminary mechanical design, trade studies and structural optimization for a high-speed pneumatic UAV catapult launcher (detailed under Projects).

Bharat Space Education Research Centre (BSERC) **June 2026 – July 2026**
Def-Space Tech Intern

– Researched pintle-type liquid bipropellant (LOX/LCH4) injector designs, deriving the foundational numerical parameters for atomization and combustion stability.
– Studied autonomous UAV architectures and defence-tech systems, applying propulsion and safety-critical design fundamentals to operational vehicle design constraints.

NHPC Ltd. – Salal Power Station **Dec 2025**
Operations & Maintenance Intern (Capital Maintenance Unit) *Reasi, J&K*

– Supported Capital Maintenance of Unit 6 (115 MW), gaining field exposure to O&M workflows and safety protocols in a high-silt river environment.
– Monitored the 400-tonne generator rotor tandem lift; analyzed Master/Slave VFD logic and industrial safety compliance procedures.
– Inspected hydraulic path integrity and documented silt-induced erosion on runners, evaluating HVOF coatings for component longevity.

PROJECTS

Movable Pintle Injector Design for LOX/LCH4 Liquid Rocket Engine **July 2026 – Present**
Independent Research – paper accepted at NCCS-30 link

– **Novel Manifold Architecture:** Conceptualized a throttleable propellant manifold applying Francis turbine spiral-casing principles to reduce circumferential swirl and uneven spray.
– **Numerical & Thermochemical Analysis:** Wrote a MATLAB script for mass-flow and geometry sizing, using NASA CEA to set a baseline design point (O/F = 3.2, 750 N thrust, 20 bar).
– **Mechanism Design:** Used literature and patent reviews (including SpaceX active-cooling and foundational 1972 patents) to design a linear actuation mechanism with dynamic seals and guide bushings.
– **Publication:** Paper on the design accepted at NCCS-30 (30th National Conference on Cryogenics and Superconductivity), to be presented in December 2026.

High-Speed Pneumatic UAV Catapult Preliminary Design **May 2026 – June 2026**
SEC Industries Private Limited (Internship Project) link

– **Parametric Sizing:** Built a 396-point MATLAB sweep (11 launch angles × 36 UAV masses, 5–49.9 kg); the governing case (20 kg at 15°) drove a 125 mm bore at a buckling factor of safety of 5.5.
– **Catalogue Component Selection:** Sized ISO 6431 cylinders and EN 286-1 accumulators (1,000 L at 10 bar; 5.93 bar working vs 8 bar max) and verified all 396 conditions against the selected hardware.
– **Architecture Trade-off:** Chose a 2:1 cable-pulley multiplier over direct drive so the launch stroke maps to a standard ISO 6431 catalogue cylinder length.
– **Benchmarking:** Reverse-engineered commercial launchers to define operational envelopes and specify UHMWPE/Teflon sliding guideways.

Leaf Vacuum Cleaner – Concept to Detailed CAD **Sept 2026 – Present**

Greeneria Renewables (Open Industry Brief) link

- **Brief to Concept:** Developed a concept in response to the director's open brief (suction-based collection of fallen leaves from roads and public spaces); an early version has been shared with the director ahead of a detailed review.
- **Architecture & Mechanisms:** Laid out the intake nozzle with stone guard, suction hose, inlet elbow, engine-driven radial impeller in a volute, discharge chute and screened hopper on a wheel-and-castor push chassis.
- **Detailed CAD:** Built a 16-part, jointed Fusion 360 assembly with assigned materials (≈222 kg), and produced orthographic, isometric, section and exploded views plus a full BOM.

Stress & Strain Contour Analysis of a Shield (FEA) **Oct 2026 – Present**

ME3103 Course Project (done individually)

- Modelling Captain America's shield in SolidWorks and running stress and strain contour analysis on it in Abaqus, handling the full CAD-to-simulation workflow alone on a group assignment.

Saral Recycle – Strategic Circular Economy Model **Sept 2025**

Business Case Study – Market Analysis, ESG Strategy

- Conceptualized a circular-economy model for Delhi's waste sector, focusing on the socio-economic integration of informal workers.
- Developed an ESG-compliant B2B framework for FMCG sourcing, targeting 1,800+ metric tons of annual waste diversion.

CASE COMPETITIONS

Break The Case '26 – Six Sigma Invoice-Error Reduction **Oct 2026**

Team Case Competition, BITS Hyderabad (Round 2) link

- **DMAIC Analysis:** Diagnosed supplier invoice errors (10,648 DPMO, 3.8σ) and built a Python pipeline that searched 25,599 solution combinations and ran 20,000 Monte Carlo scenarios.
- **Recommendation:** Phased hybrid plan projecting a 90% error cut, Rs. 162 lakh 3-year NPV and 9-month payback in the base case.

Eurogrip TORQ Challenge 4.0 (TVS Srichakra) **Oct 2026**

Case 2 – Next-Generation Tyre for EV Two-Wheelers

- Team entry with a tyre concept for electric scooters combining a range-focused compound, a zoned tread and an EV-specific tread pattern, with a planned extension to e-motorcycles.

Tata Elxsi Teliport Season 4 – VehicleGPT **Oct 2026**

Case Study 5, Team White Chocolate

- Team entry on an on-device small-language-model assistant that answers questions from a vehicle's own data, with a beta build to back the Round 1 submission.

POSITIONS OF RESPONSIBILITY

180 Degrees Consulting IIPE **Sep 2026 – Present**

President (earlier Founding Director)

- Helped found the campus chapter of the global student consultancy for non-profits and social enterprises, and ran its first-ever recruitment drive.
- As Founding Director, led marketing strategy and event execution to establish the chapter's brand presence and operating workflows.
- As President, leads the chapter's pro-bono consulting engagements, including market analysis, organisational review and brand positioning for an NGO client.

Social Welfare Secretary & NSS Student In-charge, IIPE **Aug 2025 – Aug 2026**

- Organised a blood donation camp (Feb 2026), coordinating with medical teams and logistics.
- Organised campus-wide Oral Health Awareness and Shramdaan drives, impacting 400+ students and staff.

Co-founder, Sattva Quizzing Club, IIPE **Oct 2024 – Present**

- Established the club charter and lead multi-domain quiz modules to foster analytical thinking and general awareness.

AWARDS & ACHIEVEMENTS

- **National Finalist** – Socionnovate, i5 Summit, IIM Indore (2025): selected among the top teams for social innovation.
- **Finalist** – Business and Technology Quiz, SpringFest, IIT Kharagpur (2025).
- **Paper accepted** – NCCS-30, 30th National Conference on Cryogenics and Superconductivity (movable pintle injector), presenting Dec 2026.

EDUCATION

Degree/Certificate	Institute/Board	Score	Year
B.Tech. – Mechanical Eng.	Indian Institute of Petroleum and Energy	CGPA: 7.3/10	2024–Present

Relevant Coursework: Fluid Mechanics & Multiphase Flow, Thermodynamics, Heat Transfer, Turbomachinery, Strength of Materials, Kinematics of Machines, Dynamics of Machines, Manufacturing Processes, Design of Machine Elements, Numerical Methods.